Fiscal Year:	FY 2020	**Task Last Updated:**	FY 11/14/2024
PI Name:	Lohrmann, Alexander Ph.D.		
Project Title:	Deep Space Quantum Link		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	NASA Jet Propulsion Laboratory		
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PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91011	**Congressional District:**	28
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2019	**End Date:**	09/30/2030
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	JPL 80NM0015F0506		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Quantum optical experiments in space will enable several emerging applications, ranging from cryptographic quantum communication to distribution of entanglement as a computational or measurement resource. At the same time, novel science experiments can be envisioned that exceed the sensitivities of terrestrial quantum optics. The Deep Space Quantum Link (DSQL) task explores the feasibility of these novel science experiments that can only be performed in space across large gravitational potential differences, long distances, or within special relativistic environments. The objective of the DSQL – to define and prepare a space mission to eventually implement these experiments – is split into several phases: • Science Definition Phase • Concept development		

	• Terrestrial proof-of-concepts • Mission planning Beyond these science-driven objectives, DSQL aims to mature space-quantum technology for the future global quantum internet.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New Project for FY2020
Bibliography Type:	Description: (Last Updated: 08/13/2025)